Probing the Flexibility of the Language of Thought

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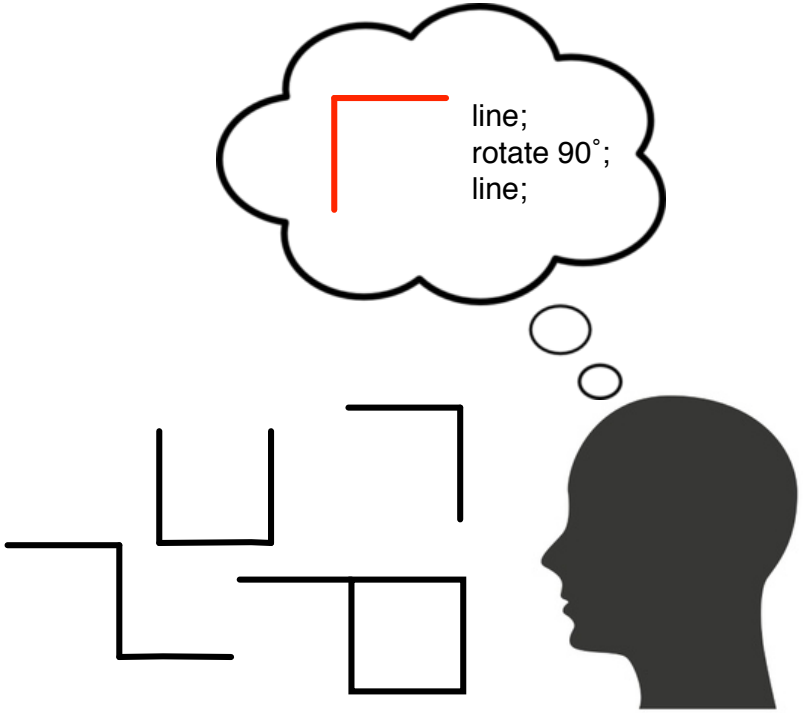
Warning

- Empirical data
- Experimentation
- The term “human” is used

(yet, there is also theory)

Language of thought



- Describe concepts relying on a LoT
- Subjective complexity of a concept for human learners is linked to the length of the minimal program that describes it
- Dependence on the primitives of the language

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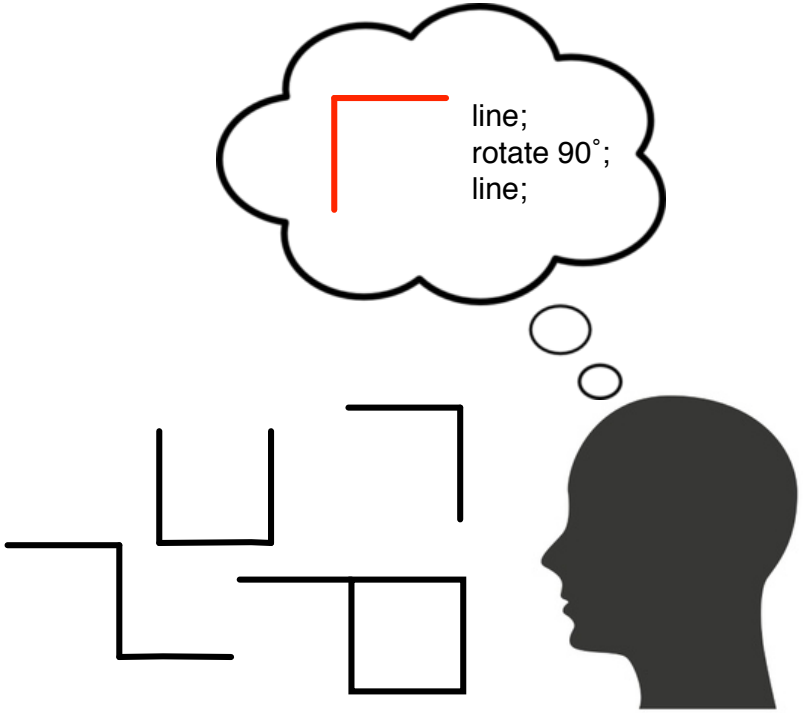
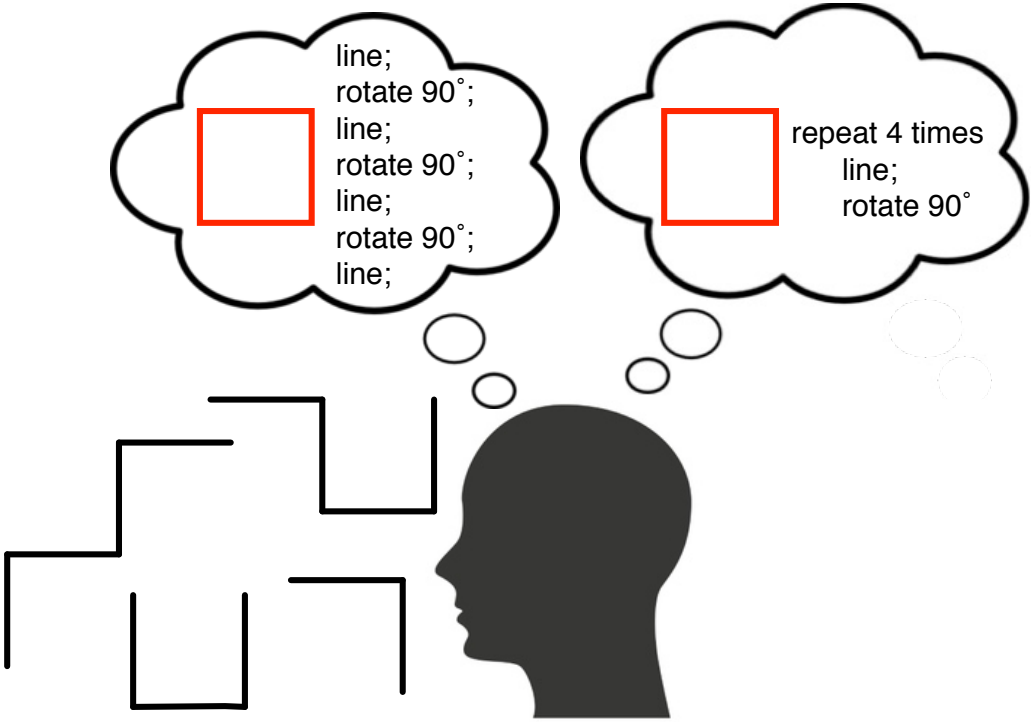


- Describe concepts relying on a LoT
- Subjective complexity of a concept for human learners is linked to the length of the minimal program that describes it
- Dependence on the primitives of the language

Are the primitives of the LoT universal? Or LoT is dynamic and shaped by experience?

Dependence with prior exposure



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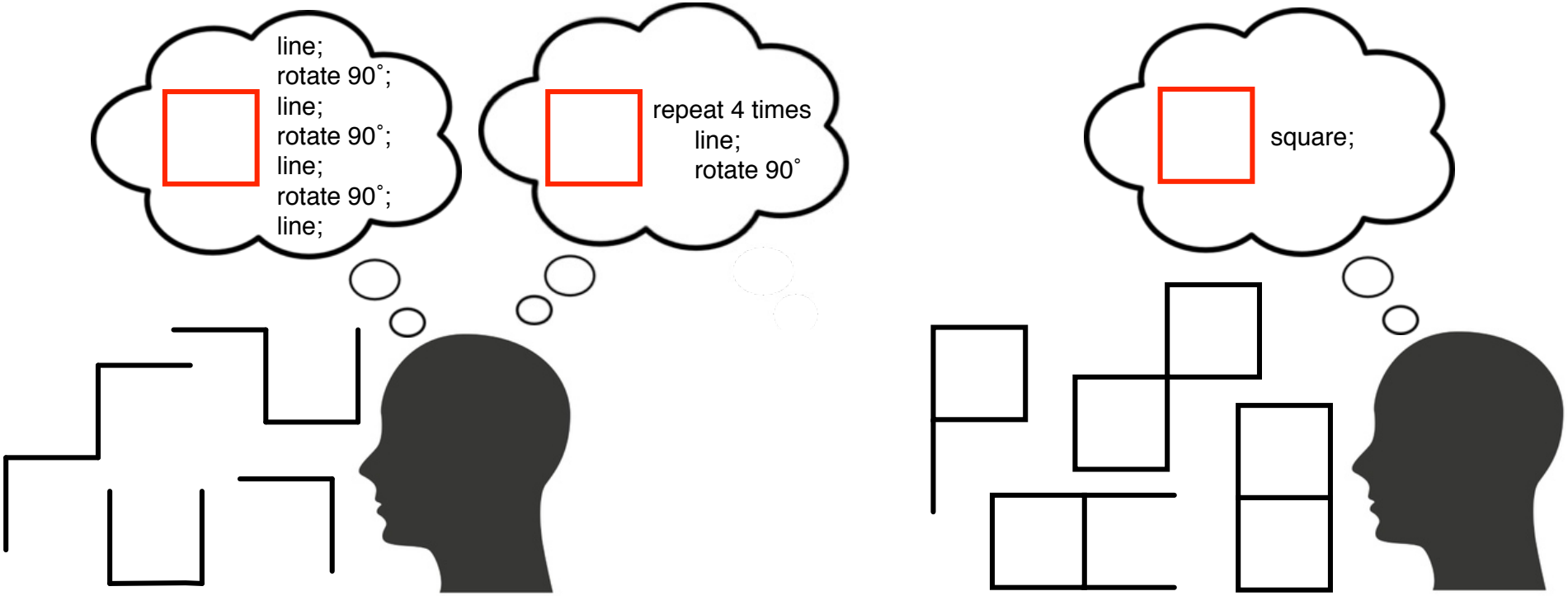

- Describe frequent concepts with new symbols
  - higher abstraction
  - release memory and processing power, thus making more complex thoughts thinkable

Understand how different tracks of experience can shape acquisition differently, and change the repertoire of a LoT

Grammars for propositional logic

Grammar Bool

$$
\begin{array}{rcl}
\textsc{start} & \rightarrow & \textsc{bool} \\
\textsc{bool} & \rightarrow & (\textsc{bool} \wedge \textsc{bool}) \\
\textsc{bool} & \rightarrow & (\textsc{bool} \vee \textsc{bool}) \\
\textsc{bool} & \rightarrow & \textsc{atom} \\
\textsc{atom} & \rightarrow & p \mid \neg p \\
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Grammar Bool$^{\oplus}$

Rules of Bool plus

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\textsc{bool} \rightarrow (\textsc{bool} \oplus \textsc{bool})
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Rules of Bool plus

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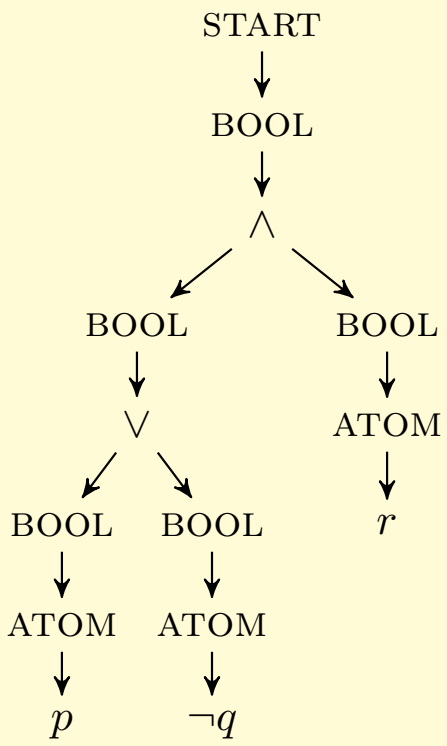
Formula

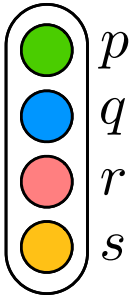
$$((p \vee \neg q) \wedge r)$$

Syntactic tree



Semantics: Boolean concepts

A formula is evaluated in a "valuation"



"trafic light"

- colored means true
- blank means false

$\neg$ negation
$\wedge$ conjunction
$\vee$ disjunction
$\oplus$ xor

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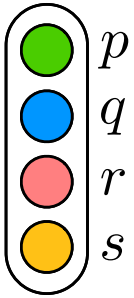

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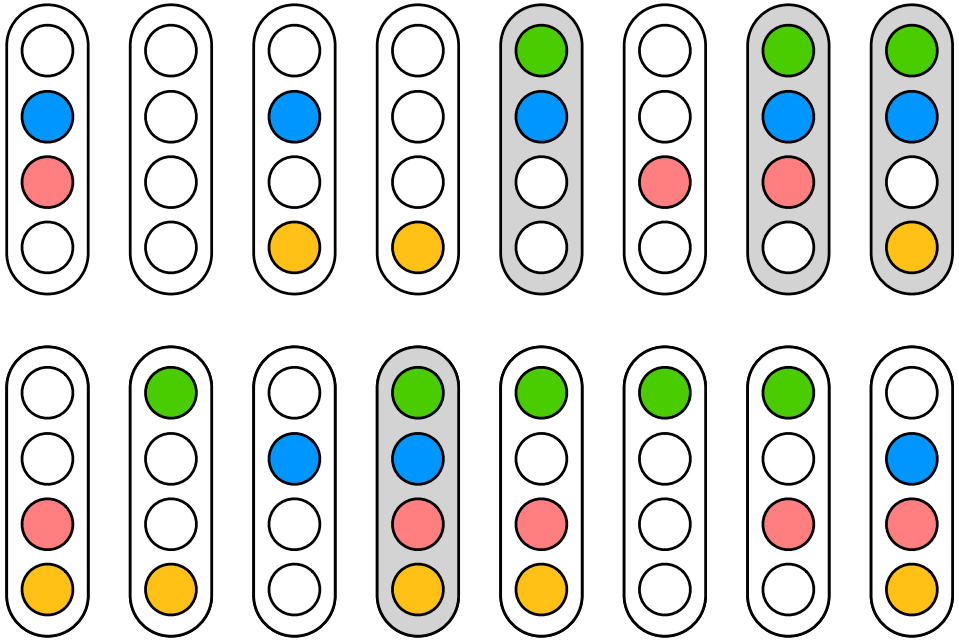
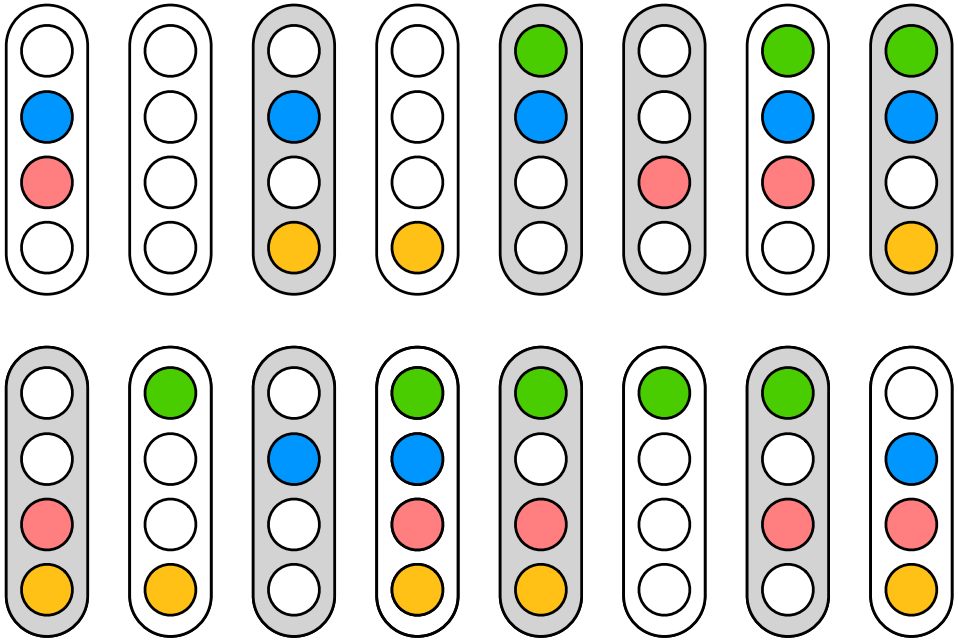
A concept C is a set of valuations.



- The semantics of a formula is the set of valuations where the formula is true
- Grayed valuations represent a concept
- We use propositional logic as language of description
- C can be described with

$$p \wedge q$$

MDL for Boolean concepts



Each concept has many descriptions

- $(\neg p \wedge q \wedge \neg r \wedge s \wedge) \vee (p \wedge q \wedge \neg r \wedge \neg s \wedge) \vee \ldots$

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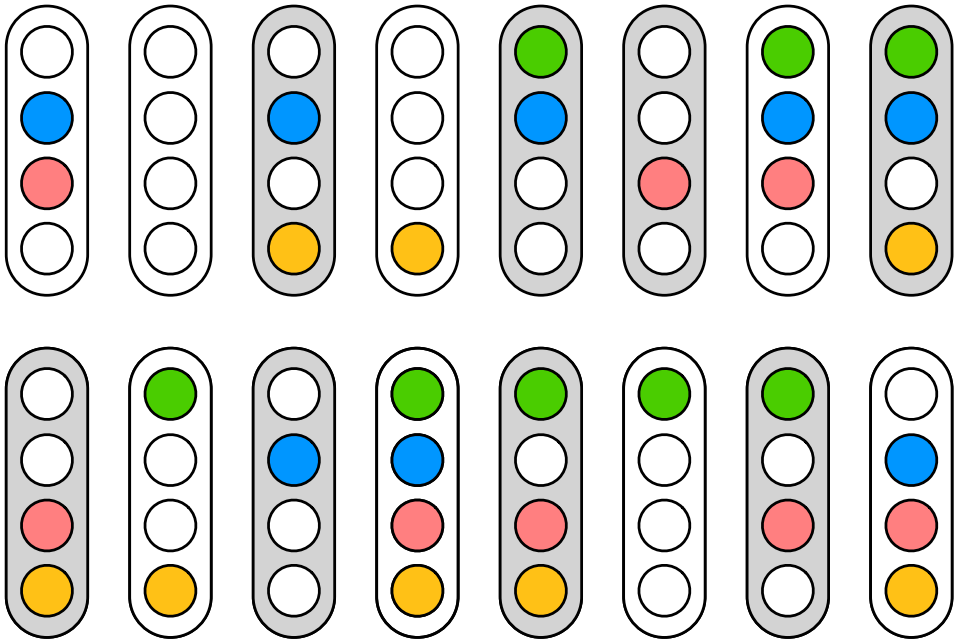

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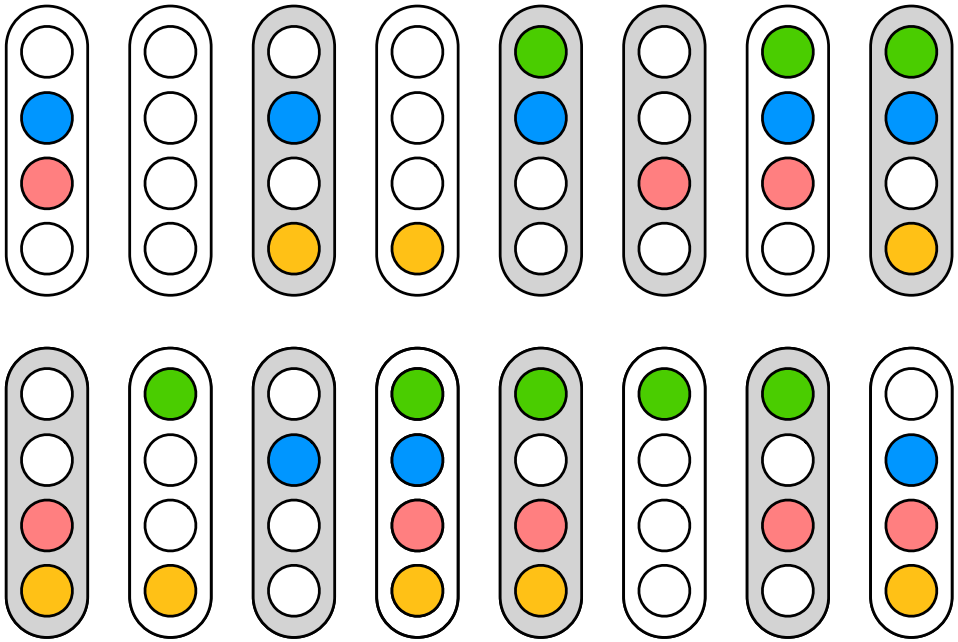

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- $(q \oplus r)$
- $\ldots$

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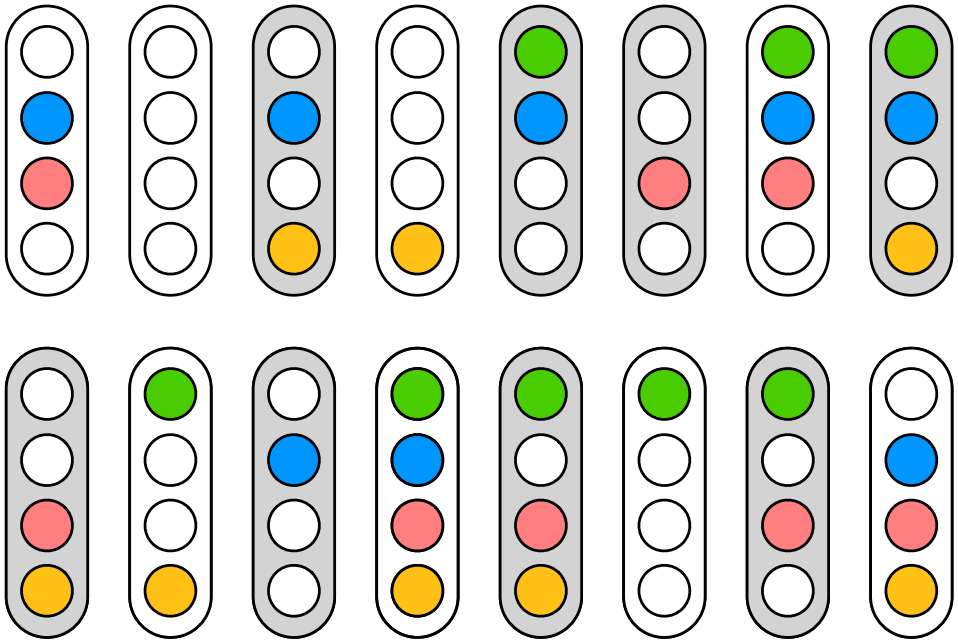

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- $(q \oplus r)$
- $\ldots$

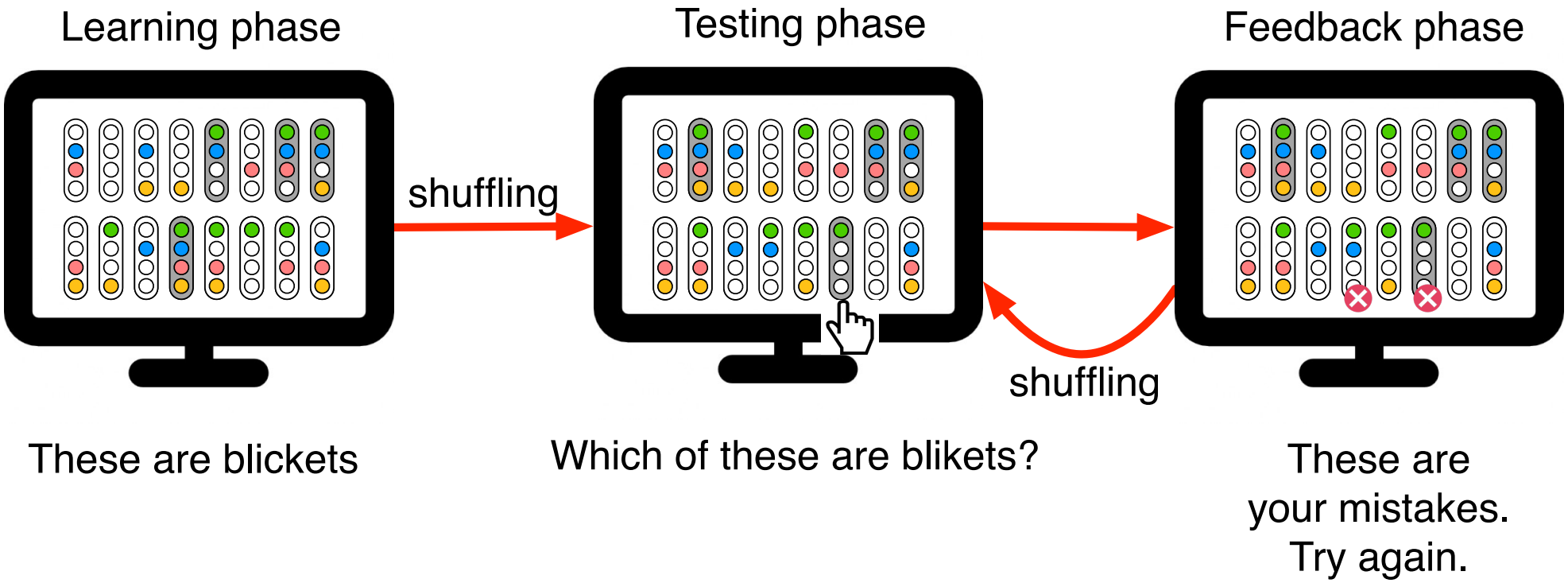
$$
\begin{aligned}
|\varphi| &= \text{number of operators and atoms of } \varphi \\
\mathsf{MDL}[\mathcal{L}](C) &= \min\{|\varphi| \mid \varphi \in \mathcal{L} \text{ and semantics of } \varphi \text{ is } C\}
\end{aligned}
$$

The minimum description length depends on the language we consider

- longer for $\mathcal{L} = \mathsf{Bool}$, shorter for $\mathcal{L} = \mathsf{Bool}^{\oplus}$

Experiment

Show 6 concepts. For each one, procedure is as follows:



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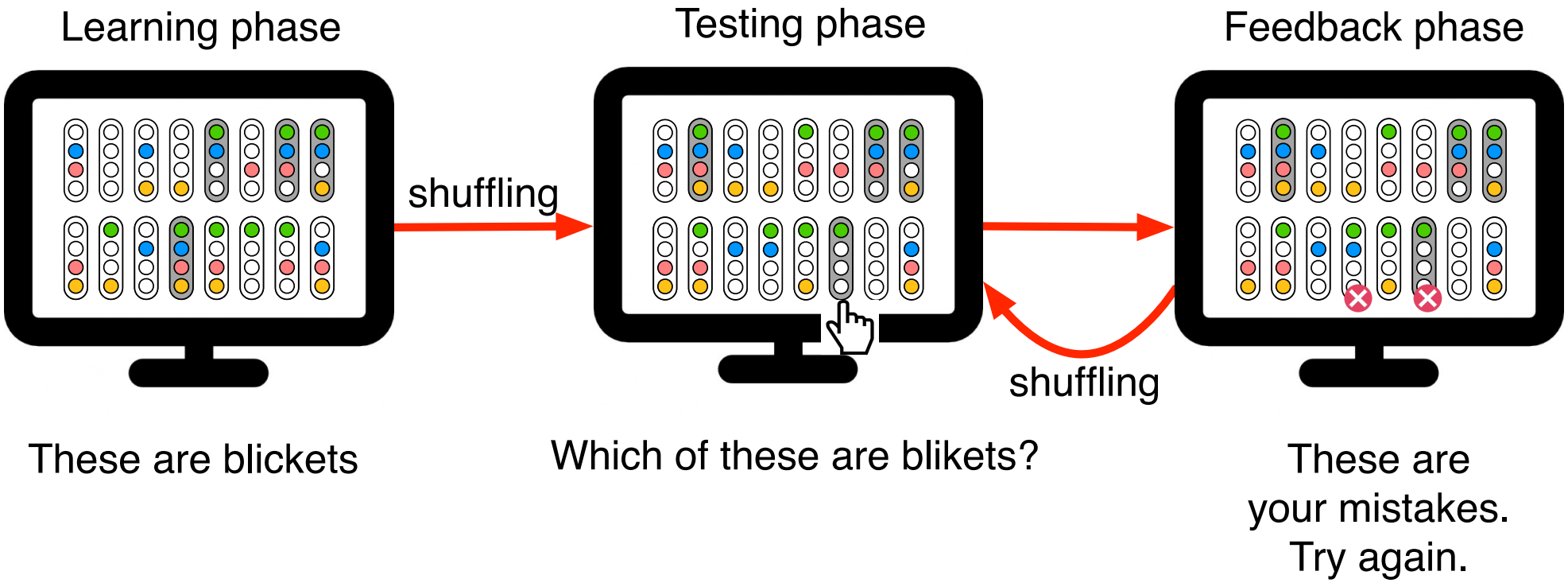

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2 groups. Each one is presented with a different sequence of concepts

- Control group: exposed to concepts easily described without $\oplus$
- Target group: exposed to concepts easily described only with $\oplus$

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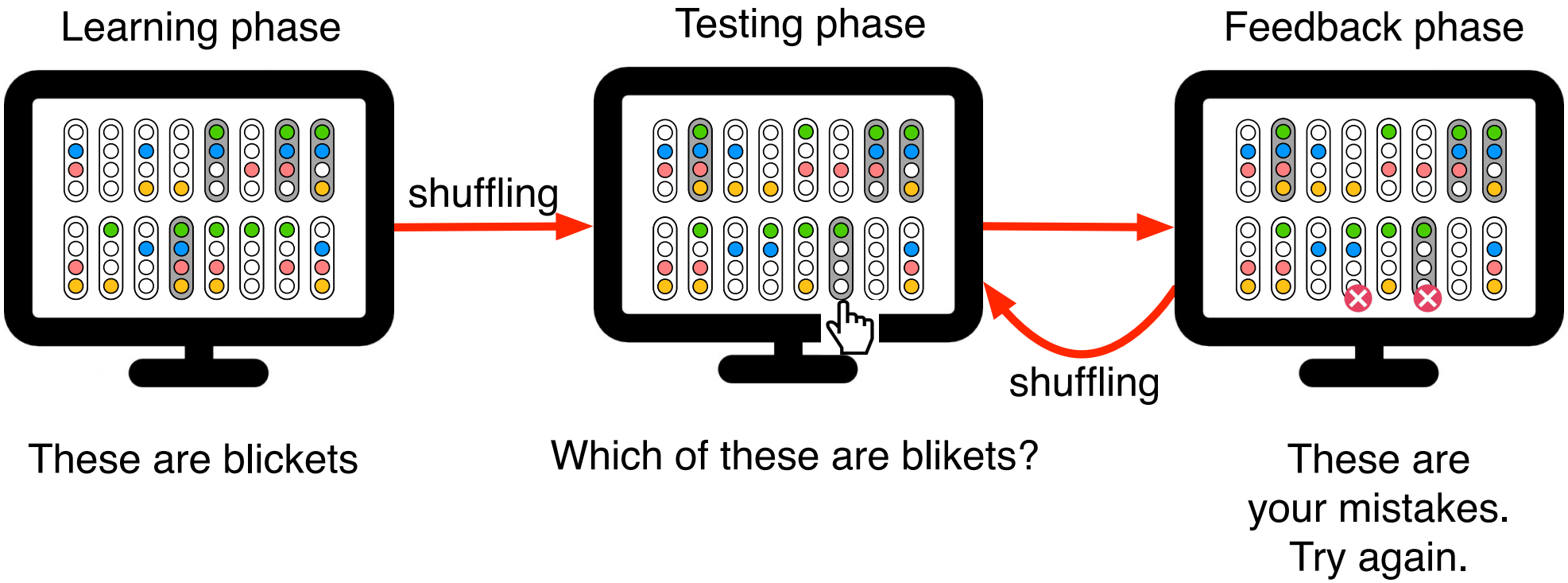

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6 concepts = 4 training + 2 test

- 4 training concepts for Control and other 4 for Target
- same 2 test concepts for both groups

Concepts for experiment

Training

Target Group (xor)			Control Group		
Concept	MDL [Bool$^{\oplus}$]	MDL [Bool]	**Concept**	MDL [Bool$^{\oplus}$]	MDL [Bool]
p	1	1	p	1	1
$p \oplus q$	3	7	$p \vee q$	3	3
$p \oplus q \oplus r$	5	19	$p \wedge q$	3	3
$r \oplus s$	3	13	$r \vee s$	3	3

Test

Concept	MDL [Bool$^{\oplus}$]	MDL [Bool]
$p \wedge (q \oplus r)$	5	9
$p \wedge (q \vee r)$	5	5

- First test concept is the target (succinctly described with $\oplus$)
- Second test concept is control ($\oplus$ does not help in compression)

Results

- We measure the subjective difficulty of concepts for participants as the total time spent for that concept in the three phases
- Analyze first test concept (same for both groups):

$$p \wedge (q \oplus r) \text{ has } \mathsf{MDL} = \begin{cases} 5 & \text{in } \mathsf{Bool}^{\oplus} \\ 9 & \text{in } \mathsf{Bool} \end{cases}$$

- To compare the performance between Target and Control groups, we calculate, for each participant, the ratio

$$\frac{\text{time spent for first (target) test concept}}{\text{time spent for both test concepts}}$$

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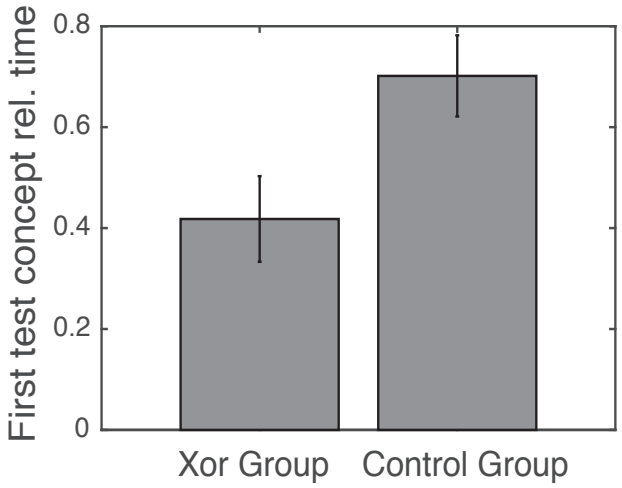

- Participants in Target group were able to go through the first concept faster than subjects in the Control group
- Subjective difficulty strongly depends on prior exposure to other concepts.

Probabilities in grammars to explain the data

Add probabilities to rules of the grammar $\mathsf{Bool}^{\oplus}$ (PCFG)

Bayes: probability of using φ to explain concept C

$$P(\varphi \mid C) \propto P(C \mid \varphi) P(\varphi)$$

$$P(C \mid \varphi) = \begin{cases} 1 & \text{semantics of } \varphi \text{ is } C \\ 0 & \text{otherwise} \end{cases}$$

Prior $P(\varphi)$ is defined by the PCFG

Subjective difficulty d_C of a concept C is proportional to the length of φ = the formula that participant uses to explain C

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We don't know φ!

Expected length of formulas used by participant under her posterior distribution

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Estimate subjective difficulty d_C

$$d_C \propto E_C(|\varphi|) + \alpha N_C$$

where

$$N_C = \begin{cases} |C| & \text{if } |C| < 8 \\ 16 - |C| & \text{otherwise} \end{cases}$$

Flexible model

$P(\varphi)$ depends on the sequence of concepts already experienced

$$P(\varphi \mid C^t) \propto P(C^t \mid \varphi) P(\varphi \mid C^1, \ldots, C^{t-1})$$

- C^t is the concept presented at trial t
- $P(\varphi \mid C^1, \ldots, C^{t-1})$ depends on the state of the PCFG at time t (which depends on the PCFG updates from trial to trial)

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The update process increases the probability of using a certain rule of $\mathsf{Bool}^{\oplus}$ according to how useful this rule was to compress concepts previously learned. Formalization as in

Piantadosi, Tenenbaum, & Goodman. The logical primitives of thought: Empirical foundations for compositional cognitive models. Psychol. Rev., 2016

- Use a single Dirichlet-multinomial for the set of rule expansions
- Using a rule once will substantially increase its probability of being reused in the future.

Update

$$P(\varphi \mid C^t) \propto P(C^t \mid \varphi)\, \underbrace{P(\varphi \mid C^1, \ldots, C^{t-1})}_{P(\varphi \mid \mathbf{D}^t)}$$

- $\mathbf{D}^t = (D_1^t, \ldots, D_n^t)$ is a vector or reals containing the parameters of Dirichlet at step t
  - D_i^t corresponds to the ith rule of $\mathsf{Bool}^{\oplus}$

$$\mathbf{D}_i^0 = \begin{cases} 10^{-4} & \text{the } i\text{th rule is } \textsc{bool} \to (\textsc{bool} \oplus \textsc{bool}) \\ 1 & \text{otherwise} \end{cases}$$

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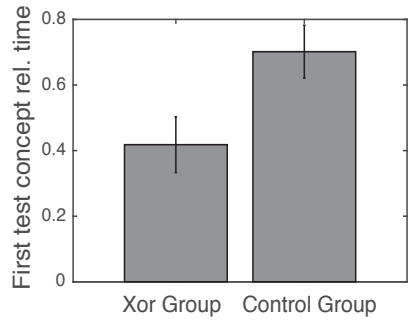
Update rule

$$D_i^{t+1} = D_i^t + \sum_{\substack{\psi \text{ s.t.} \\ \text{semantics} \\ \text{of } \psi \text{ is } C^t}} P(\psi \mid \mathbf{D}^t)\; M_i(\psi)$$

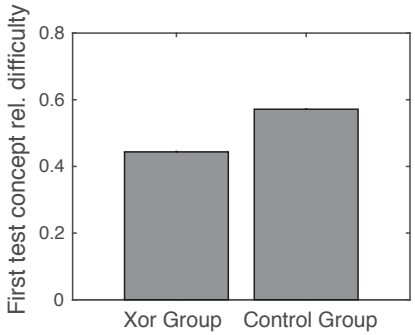
where $M_i(\psi) = \#$times rule i is used in generating ψ

Model results

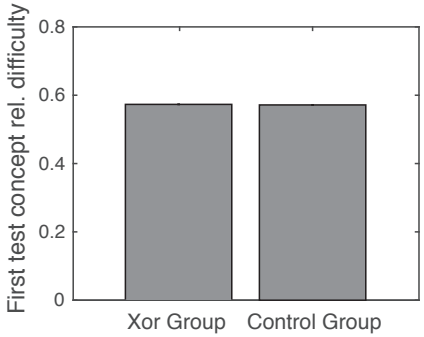
Humans



Model



Inflexible model
(no updates)



Performance
Control vs. Target groups

Model results

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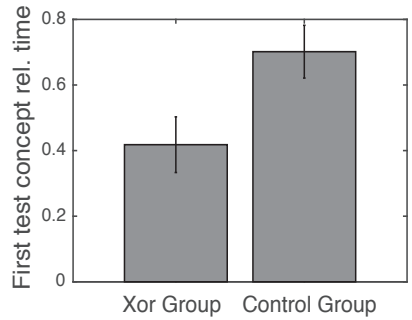

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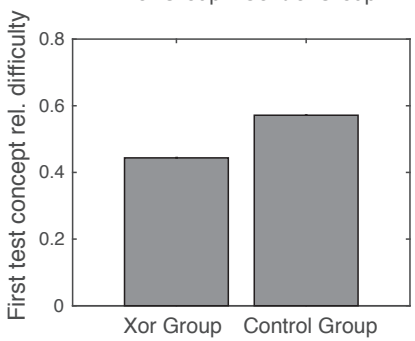

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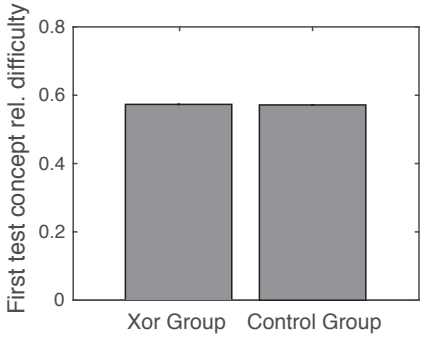

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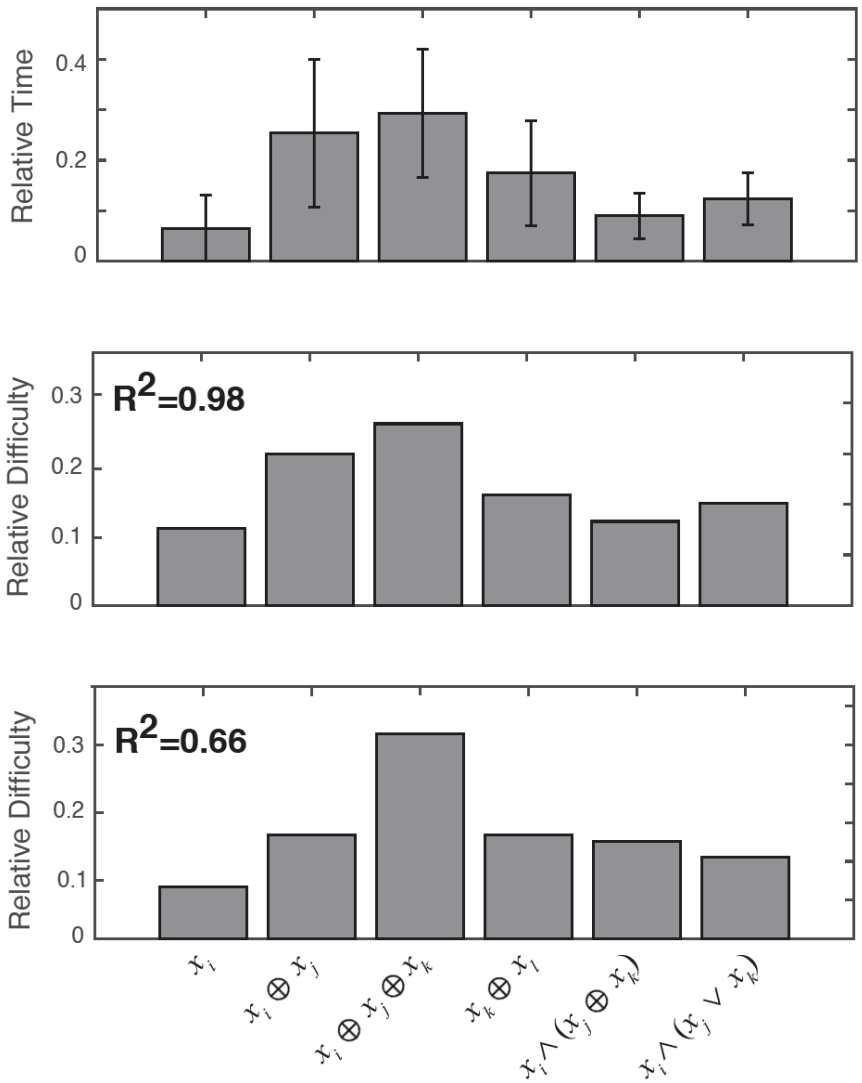

Predicted difficulty
in Target group

Temporal logic

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Grammar $\mathsf{Temp}^{\mathsf{F}}$

Rules of Temp plus

$$
\begin{array}{lcl}
\textsc{bool} & \rightarrow & \mathsf{F}\ \textsc{bool} \\
\textsc{bool} & \rightarrow & \mathsf{G}\ \textsc{bool} \\
\textsc{atom} & \rightarrow & r \mid \neg r
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Grammar Temp^{F}

Rules of Temp plus

BOOL	$\rightarrow$	F BOOL
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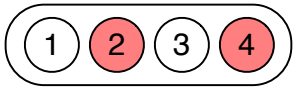
A formula is evaluated in a certain point $i \in \{1, 2, 3, 4\}$ of a "valuation"

$v =$ 

$v, i \models \varphi$ means "φ is true at i under valuation v"

- Semantics of Temp is just as Bool:

Example

$v, i \models 2 \wedge 4 \quad v, i \not\models 1 \vee 3 \quad$ irrespective of i

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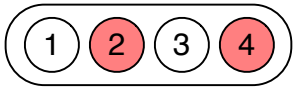
$v =$ 

$v, i \models \varphi$ means "φ is true at i under valuation v"

- Semantics of Temp is just as Bool:

Example

$v, i \models 2 \wedge 4 \quad v, i \not\models 1 \vee 3 \quad$ irrespective of i

- Semantics of Temp^{F} adds temporal modality

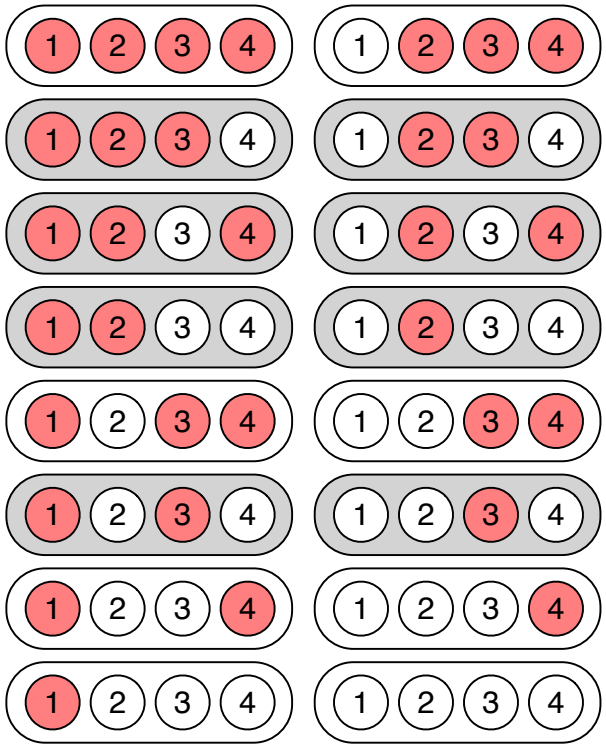
$$
\begin{array}{rcl}
v, i \models r & \text{iff} & i \text{ is red} \\
v, i \models \mathsf{F}\varphi & \text{iff} & \text{there is } j > i \text{ s.t. } v, j \models \varphi \\
v, i \models \mathsf{G}\varphi & \text{iff} & \text{for all } j > i,\ v, j \models \varphi
\end{array}
$$

Example

$v, 1 \models \mathsf{F}r \qquad v, 1 \models \mathsf{F}\neg r \qquad v, 2 \not\models \mathsf{F}(r \wedge \neg 4)$
$v, 1 \models \mathsf{F}(r \wedge \mathsf{F}\neg r)$

Semantics: temporal concept

A concept C is a set of valuations.



- The semantics of a formula φ is

$$\{v \mid v, 1 \models \varphi\}$$

  i.e. the set of valuations where the formula is true when evaluated at 1.
- Grayed valuations represent a concept
- We use temporal logic as language of description
- C can be described with

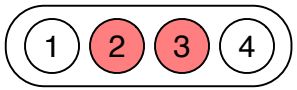
$$\mathsf{F}(r \wedge \mathsf{F}\neg r)$$

  whose meaning is

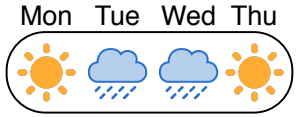
  in the future there is a red point, and in the future of that point there is a non-red point

Experiment

- Same as before
- More friendly context
- *This fruit only grows in certain types of "weeks"*



$$
\begin{aligned}
r &= \textit{red} \\
\neg r &= \textit{white} \\
1 &= 1 \textit{ is red} \\
2 &= 2 \textit{ is red} \\
3 &= 3 \textit{ is red} \\
4 &= 4 \textit{ is red}
\end{aligned}
$$



$$
\begin{aligned}
r &= \textit{rains} \\
\neg r &= \textit{sunny} \\
1 &= \textit{rain on Monday} \\
2 &= \textit{rain on Tuesday} \\
3 &= \textit{rain on Wednesday} \\
4 &= \textit{rain on Thursday}
\end{aligned}
$$

Concepts for experiment

Training

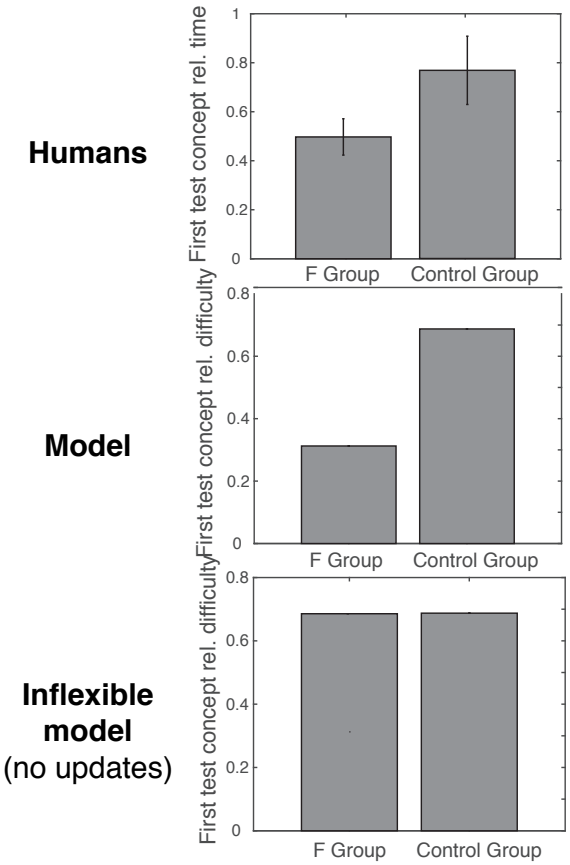
Target Group (F)			Control Group		
Concept	MDL [$\mathsf{Temp}^{\mathsf{F}}$]	MDL [Temp]	**Concept**	MDL [$\mathsf{Temp}^{\mathsf{F}}$]	MDL [Temp]
2	1	1	2	1	1
$\mathsf{F}(r \wedge \mathsf{F}(\neg r \wedge \mathsf{F}r))$	8	11	$1 \vee \neg 2$	3	3
$\mathsf{F}(r \wedge \mathsf{F}(\mathsf{F}\neg r)$	6	9	$1 \vee (\neg 3 \wedge 4)$	5	5

Test

Concept	MDL [$\mathsf{Temp}^{\mathsf{F}}$]	MDL [Temp]
$\mathsf{F}(r \wedge \mathsf{F}\neg r)$	5	11
$1 \wedge (2 \vee \neg 4)$	5	5

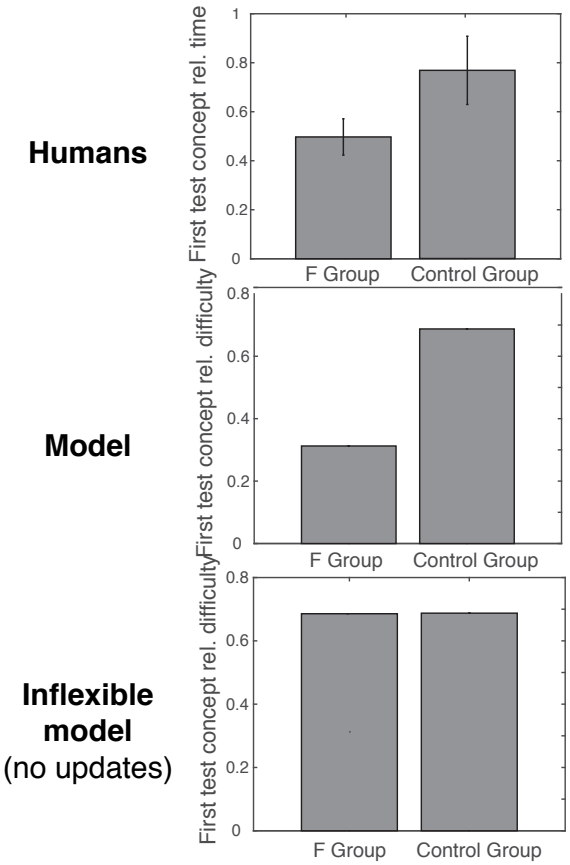
- First test concept is the target (succinctly described with F)
- Second test concept is control (F or G do not help in compression)

Model results

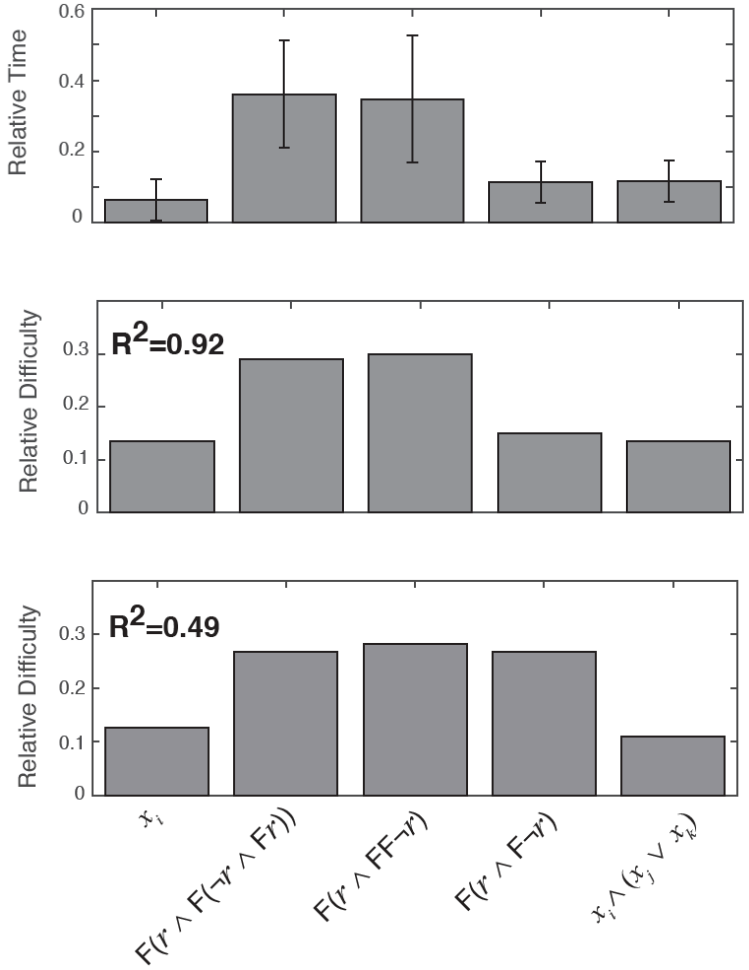


Performance
Control vs. Target groups

Model results

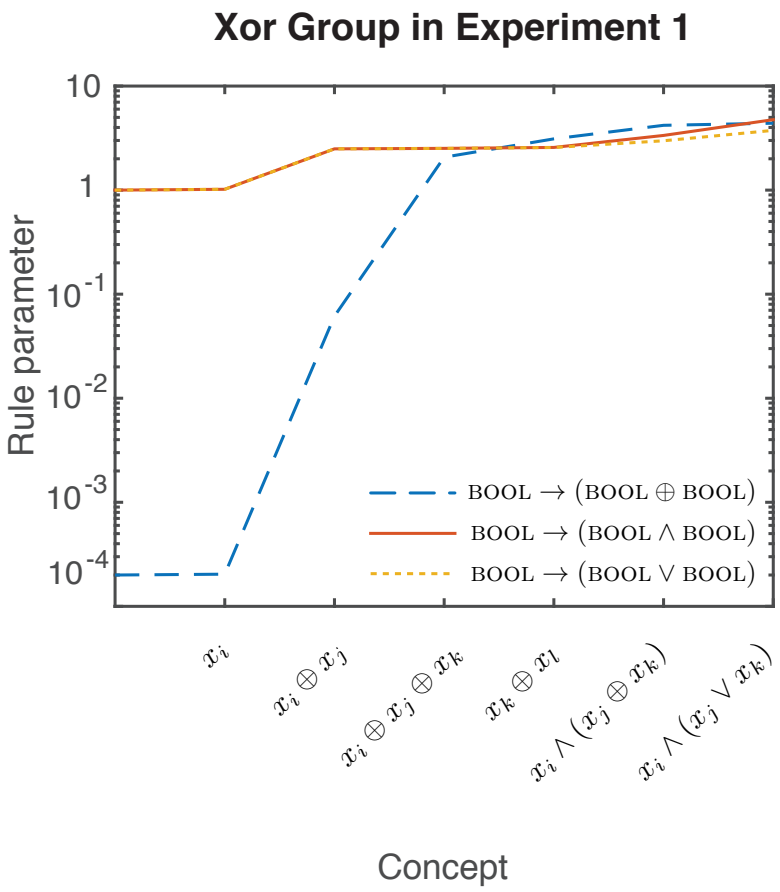


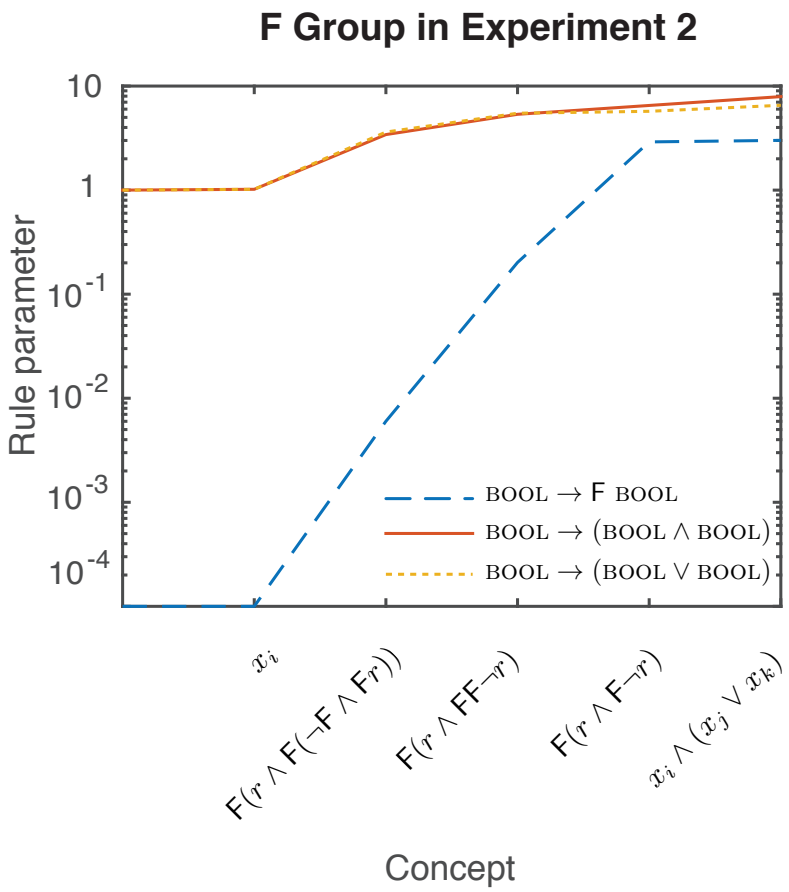
Performance
Control vs. Target groups



Predicted difficulty
in Target group

Evolution of Dirichlet parameters





Conclusions

- Relative difficulty in learning different concepts strongly depends on the exposition to prior concepts in the domain.
- Time taken to learn a concept is accurately predicted by a Bayesian updated in a PCFG
  - includes some latent compiled functions, which are a priori possible to use, but with vanishing probability
  - as they become useful, their usage probability increases, reducing future learning times.
- One cannot simply ignore the choice of concepts in experiments which aim at uncovering the specifics of the LoT
  - when a specific grammar beats a second one at fitting some experimental data, what we may be seeing is an effect of prior experience (including from the experiment itself), more than an intrinsic feature of the LoT

Thank you!